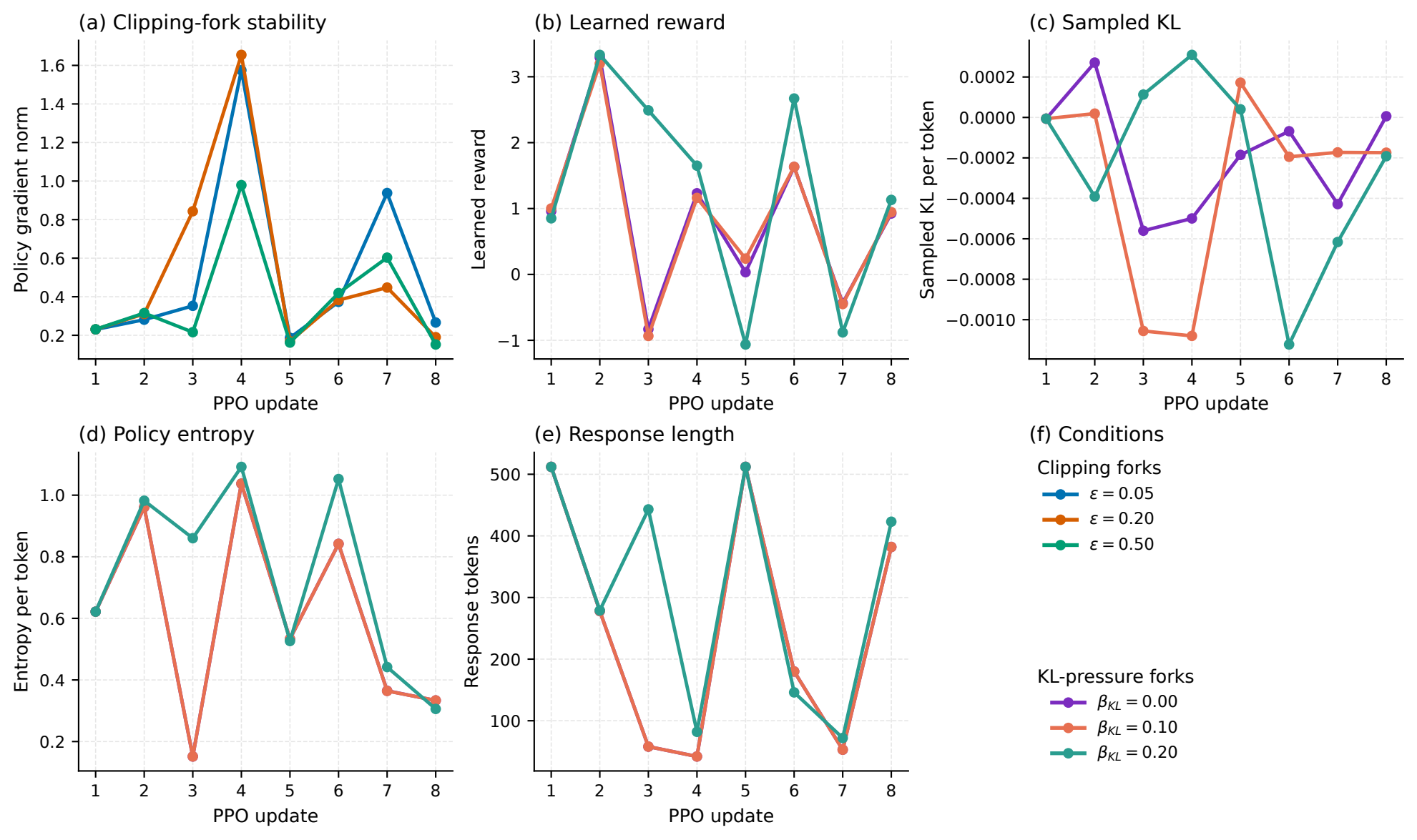
(a) Clipping-fork stability
Policy gradient norm
PPO update
(b) Learned reward
Learned reward
PPO update
(c) Sampled KL
Sampled KL per token
PPO update
(d) Policy entropy
Entropy per token
PPO update
(e) Response length
Response tokens
PPO update
(f) Conditions
Clipping forks
$\varepsilon = 0.05$
$\varepsilon = 0.20$
$\varepsilon = 0.50$
KL-pressure forks
$\beta_{KL} = 0.00$
$\beta_{KL} = 0.10$
$\beta_{KL} = 0.20$